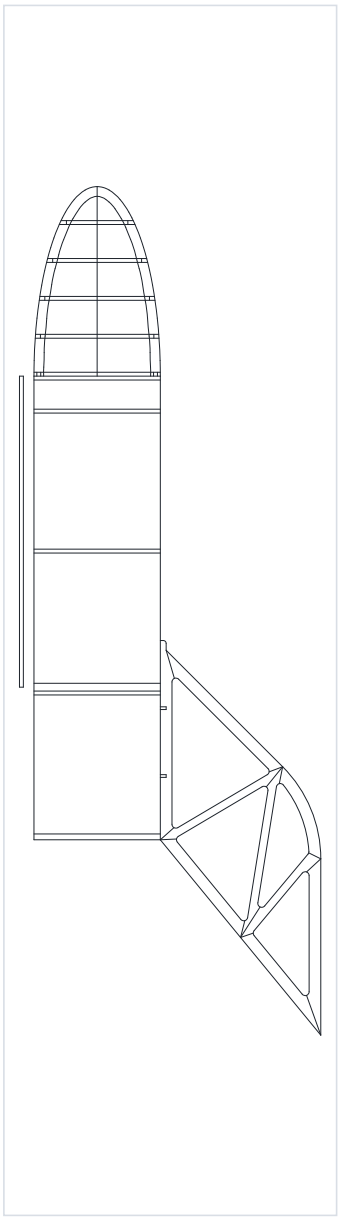
Zhang Mingchang
Rocket Design Portfolio

Hardware design and fabrication, solid-fuel rocket group, Hebei University of Science and Technology aeromodelling team (2020–2022)

01 The rocket



Assembled rocket with parachute, Nov 2020



CAD side view of the same rocket (65 mm diameter, 336 mm long)

MY ROLE

Hardware
Mainly responsible for the structural design, drawings and fabrication of the whole rocket.

Every airframe part
Nose cone, fins, body tube and recovery system.

Design process
Overall layout in OpenRocket, checking centre of mass and centre of pressure for stability. Details modelled in SolidWorks.

Recovery system
Designed the piston ejection that replaced the side hatch (page 2).

Group leader
Led the rocket group in my second year.

Team split
A teammate handled the software.

HOW THE PARTS WERE MADE

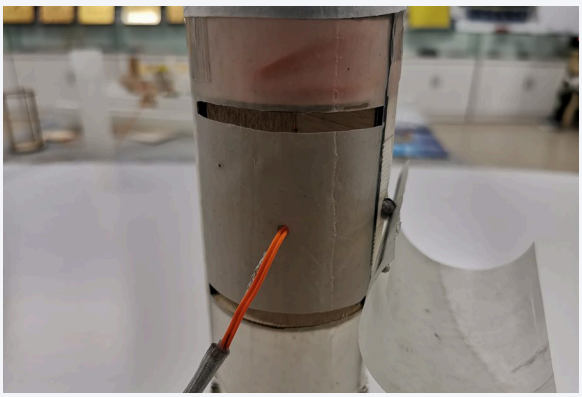
01	02	03	04	05
OpenRocket layout	**SolidWorks detail**	**DXF to laser cutter**	**Composite tube + carbon frame**	**Assembly and flight test**

Materials: balsa and plywood (laser-cut), carbon rods, carbon-fibre reinforcement, fibreglass-and-resin tube, 3D-printed parts.

02 Recovery system: piston ejection

My main contribution. The old design failed about half the time. I replaced it with a piston, proved it on the ground, and the team adopted it.

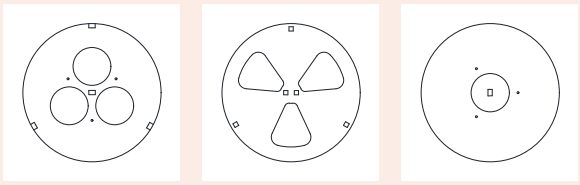
BEFORE: SIDE HATCH



The parachute left through a side hatch. The canopy caught on the rocket body, so it opened only about 1 time in 2.

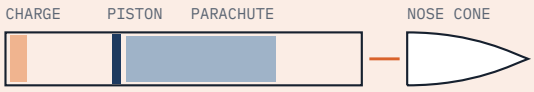
Side hatch on the earlier body tube, Nov 2020

AFTER: PISTON EJECTION



My laser-cut drawings of the piston parts, Sep 2020

Gas from a small ejection charge drives a piston that pushes the nose cone and the parachute out of the tube together, so the canopy has nothing to snag on. The charge size was found by testing.



RESULTS

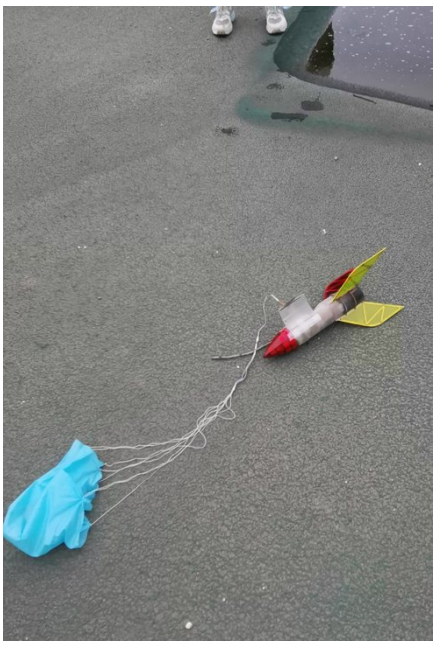
Ground test
Proved on the ground first, then adopted by the team

50+
test and practice launches

≈ 9 / 10
parachute openings after the change

4th
national, team event, CADC 2021

None of the ten cohorts before us had tried this structure. Later teams kept using it.



Landed under its parachute, test flight, Jul 2021



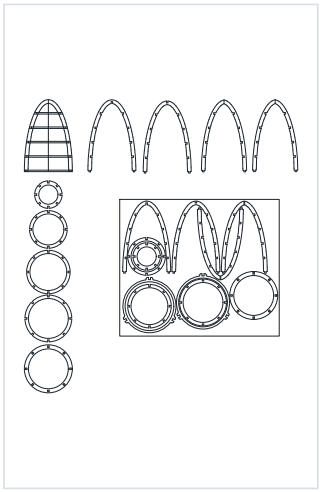
Recovered rocket after a test flight, Jul 2021



Launch at the CADC 2021 national final, Fuxin

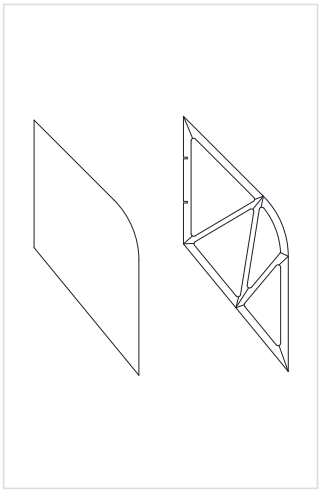
03 Fabrication: from drawing to part

Parts were drawn in CAD, exported as DXF and cut on a laser cutter. Each pair below shows a drawing and the part made from it.





Nose cone: laser-cut rib set, before skinning

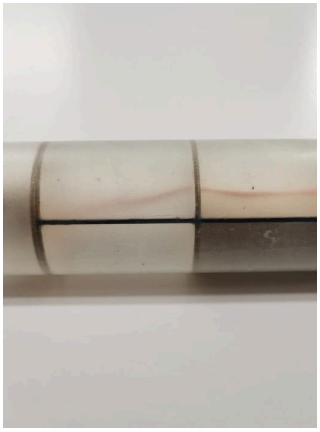




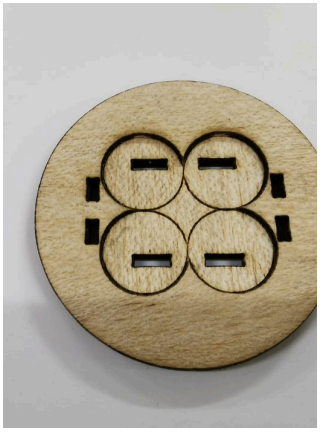
Fins: balsa truss frame from the cutting drawing



Fin frame after covering



Fibreglass-and-resin body tube



Laser-cut motor-mount bulkhead



Tube section with bulkhead fitted

AS GROUP LEADER: TARGET-ALTITUDE EVENT

The event was scored on how close the rocket's highest point came to 40 m. Before training began, I redesigned the internal structure and drew and built it myself. Its first test flight reached 38 m, and it set the direction for the group's competition model that year.

ALSO

Third Prize, University Aerospace Innovation Challenge 2021 (university round), with a model rocket where I did everything except the software.

VIDEOS: SCAN THE CODE OR CLICK THE BOX

VIDEO 1

Test flight: landing under the parachute

youtube.com/shorts/f59oAd91Sxc

VIDEO 2

CADC 2021 national final: launch

youtube.com/shorts/V23yIQ5HWlk

VIDEO 3

Test flight, launch to recovery

youtube.com/shorts/x3F7QiUBE5Y